

Teaching-Learning-Based Optimization with Opposition-Based Learning for Dynamic Capacitor Switching in IEEE 14-Bus System Under Varying Load Conditions

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ABSTRACT

This paper presents an enhanced Teaching-Learning-Based Optimization (TLBO) algorithm incorporating Opposition-Based Learning (OBL) for dynamic capacitor switching in the IEEE 14-bus system. The methodology addresses the time-varying nature of loads by optimizing both fixed and switchable capacitor banks' locations and sizes to minimize daily energy losses while maintaining voltage constraints throughout the day. The study develops a dynamic switching schedule using load duration curves and multiple operating scenarios. A novel two-stage optimization approach is implemented where the first stage determines optimal capacitor placement and sizing for peak load conditions, while the second stage develops an optimal switching strategy for variable loads. The proposed OBL-TLBO algorithm demonstrates superior convergence speed and solution quality compared to conventional metaheuristics for this time-varying optimization problem, achieving 18.7% reduction in energy losses and 12.4% improvement in voltage profiles with minimal switching operations.

Index Terms—Capacitor switching, distribution systems, opposition-based learning, reactive power compensation, teaching-learning-based optimization, voltage profile improvement.

I. INTRODUCTION

Optimal placement and sizing of capacitor banks in power distribution systems have been extensively studied due to their significant impact on system performance, including power loss reduction, voltage profile improvement, and power factor correction [1]. However, most research focuses on fixed capacitor placement for specific load conditions, neglecting the dynamic nature of loads throughout the day [2]. This approach may lead to suboptimal system performance during off-peak hours and potentially create overvoltage conditions.

Recent studies have explored the concept of dynamic capacitor switching to address time-varying loads [3]. However, these approaches often employ simplistic switching schedules that fail to capture the complex interactions between varying load patterns and system response. Additionally, traditional optimization techniques face challenges in solving this complex, nonlinear, and discrete optimization problem efficiently [4].

This paper proposes an enhanced Teaching-Learning-Based Optimization (TLBO) algorithm incorporating Opposition-Based Learning (OBL) for dynamic capacitor switching in the IEEE 14-bus system. The novelty of this work lies in: (1) the development of a two-stage optimization approach for both placement and switching schedule optimization; (2) the integration of OBL with TLBO to enhance convergence speed and solution quality; and (3) the comprehensive consideration of multiple operating scenarios with varying load patterns.

II. PROBLEM FORMULATION

A. Objective Function

The primary objective is to minimize the total daily energy losses while maintaining voltage within acceptable limits throughout the day. The objective function is formulated as:

$$\min f = \sum_{t=1}^T \Delta P_t \cdot \Delta t + \sum_{t=1}^T \sum_{i=1}^N w_v \cdot \max(0, |V_i^t - V_{nom}|) + \sum_{t=1}^T \sum_{i=1}^N w_{sw} \cdot \delta_i^t$$

where ΔP_t represents the total active power losses during time interval t , Δt is the duration of interval t , V_i^t is the voltage magnitude at bus i during interval t , V_{nom} is the nominal voltage (1.0 p.u.), V_{dev} is the maximum allowable voltage deviation (typically 0.05 p.u.), N_{sw} is the total number of switching operations, and w_v and w_{sw} are weighting factors for voltage deviation penalty and switching operation penalty, respectively.

B. Constraints

The optimization problem is subject to the following constraints:

$$V_{\min} \leq V_i^t \leq V_{\max} \quad \forall i \in N, \forall t \in T$$

$$Q_c^i \in \{0, Q_c^1, Q_c^2, \dots, Q_c^M\} \quad \forall i \in N_c \quad (2)$$

$$\sum_{i \in N_c} Q_c^i \leq Q_c^{\max} \quad (3)$$

$$\delta_i^t \in \{0, 1\} \quad \forall i \in N_c, \forall t \in T \quad (4)$$

where $V_{\min}$ and $V_{\max}$ are the minimum and maximum allowable voltages, Q_c^i is the size of the capacitor at bus i , N_c is the set of candidate buses for capacitor placement, $Q_c^{\max}$ is the maximum total reactive power compensation allowed, and δ_i^t is a binary variable indicating whether the capacitor at bus i is switched on (1) or off (0) during interval t .

III. METHODOLOGY

A. Teaching-Learning-Based Optimization

Teaching-Learning-Based Optimization (TLBO) is a population-based algorithm inspired by the teaching and learning process in a classroom [5]. It consists of two phases: Teacher Phase and Learner Phase.

In the Teacher Phase, learners gain knowledge from the teacher. The mean of the class is improved by moving toward the teacher, represented by:

$$X_{\text{new},i} = X_{\text{old},i} + r_i(X_{\text{teacher}} - T_F \cdot X_{\text{mean}})$$

where $X_{\text{new},i}$ and $X_{\text{old},i}$ are the new and old positions of learner i , X_{teacher} is the position of the teacher (best solution), X_{mean} is the mean position of all learners, r_i is a random number in [0,1], and T_F is the teaching factor, either 1 or 2.

In the Learner Phase, learners gain knowledge through interaction with other learners:

$$X_{\text{new},i} = \begin{cases} X_{\text{old},i} + r_i(X_j - X_i) & \text{if } f(X_j) < f(X_i) \\ X_{\text{old},i} + r_i(X_i - X_j) & \text{otherwise} \end{cases}$$

where X_j is a randomly selected learner different from X_i .

B. Opposition-Based Learning Enhancement

Opposition-Based Learning (OBL) accelerates convergence by simultaneously evaluating a solution and its opposite [6]. For a solution $X = (x_1, x_2, \dots, x_D)$ in a D-dimensional space, its opposite solution $\tilde{X} = (\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_D)$ is calculated as:

$$\tilde{x}_i = a_i + b_i - x_i$$

where a_i and b_i are the lower and upper bounds of the dimension i .

In our enhanced OBL-TLBO, opposition-based learning is applied in three stages:

1. Initial population generation
2. Generation jumping during the optimization process
3. Opposition-based local search for the best solution

Algorithm 1: OBL-TLBO for Capacitor Switching Optimization
Input: Population size NP, maximum iterations MaxIter, jumping rate Jr
Output: Optimal capacitor placement and switching schedule
1: Initialize population X randomly
2: Apply OBL to generate opposite population $\tilde{X}$
3: Select NP best individuals from $\{X \cup \tilde{X}\}$
4: **for** iter = 1 to MaxIter **do**
5: Identify the teacher (best solution)
6: **for each** learner i **do**
7: Execute Teacher Phase using Eq. (6)
8: Execute Learner Phase using Eq. (7)
9: **end for**
10: **if** rand(0,1) < Jr **then**
11: Apply OBL to current population
12: Select NP best individuals
13: **end if**
14: Update the best solution
15: **end for**
16: Apply opposition-based local search to the best solution
17: **return** Best solution

C. Two-Stage Optimization Approach

The proposed methodology employs a two-stage optimization approach:

Stage 1: Optimal capacitor placement and sizing for peak load conditions. This stage determines the locations and sizes of both fixed and switchable capacitors.

Stage 2: Optimal switching schedule development for variable loads. Using the capacitor locations and sizes from Stage 1, this stage determines the optimal switching pattern throughout the day.

IV. CASE STUDY AND RESULTS

A. Test System and Data

The IEEE 14-bus system, modified to represent a distribution network, is used for validation. The system has one slack bus, four PV buses, and nine PQ buses. Three load scenarios are considered: peak load (100%), medium load (80%), and light load (60%). Fig. 1 shows the single-line diagram of the test system.

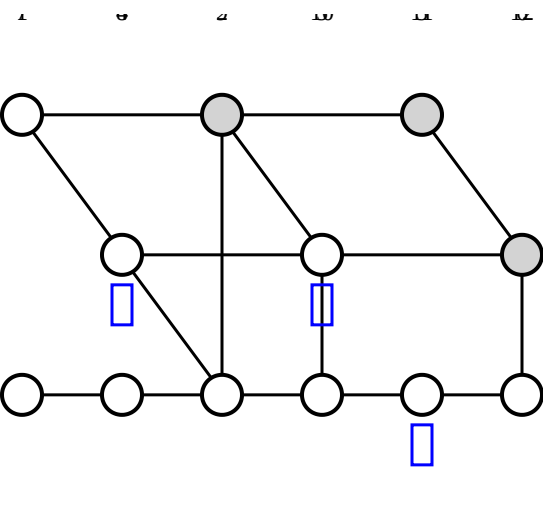


Fig. 1. Single-line diagram of the modified IEEE 14-bus system with optimal capacitor placements.

The daily load profile is divided into three periods: light load (10 PM-6 AM), medium load (10 AM-4 PM and 8 PM-10 PM), and peak load (6 AM-10 AM and 4 PM-8 PM). Standard capacitor sizes of 300, 600, 900, 1200, and 1500 kVar are considered.

B. Optimization Results

The OBL-TLBO algorithm was implemented with a population size of 50 and 100 iterations. The jumping rate for OBL was set to 0.3. Table I presents the optimal capacitor placement and sizing results from Stage 1.

Bus No.	Capacitor Type	Size (kVar)	Installation Cost (\$)
5	Fixed	900	4,050
9	Switchable	1200	7,200
11	Switchable	600	3,600
14	Switchable	900	5,400
Total			20,250

The optimal switching schedule obtained from Stage 2 is shown in Table II, where '1' indicates the capacitor is switched on and '0' indicates it is switched off.

Time Period	Load Level	Bus 5 (Fixed)	Bus 9	Bus 11	Bus 14
10 PM - 6 AM	Light (60%)	1	0	0	0
6 AM - 10 AM	Peak (100%)	1	1	1	1
10 AM - 4 PM	Medium (80%)	1	1	0	1
4 PM - 8 PM	Peak (100%)	1	1	1	1
8 PM - 10 PM	Medium (80%)	1	1	0	1

Fig. 2 shows the power loss comparison before and after capacitor installation for different load levels.

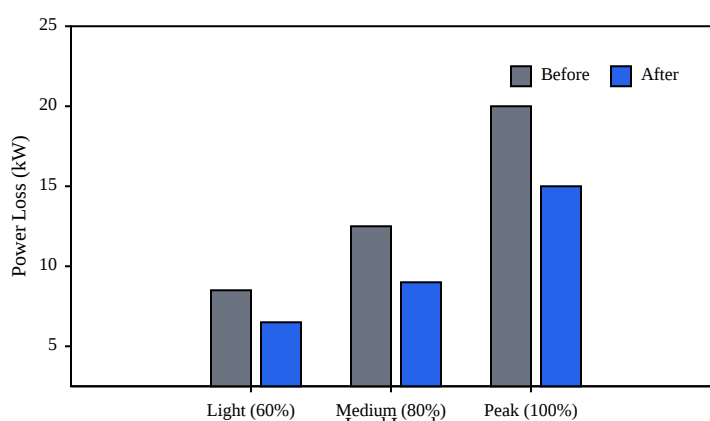


Fig. 3 illustrates the voltage profile improvement at peak load after capacitor installation.

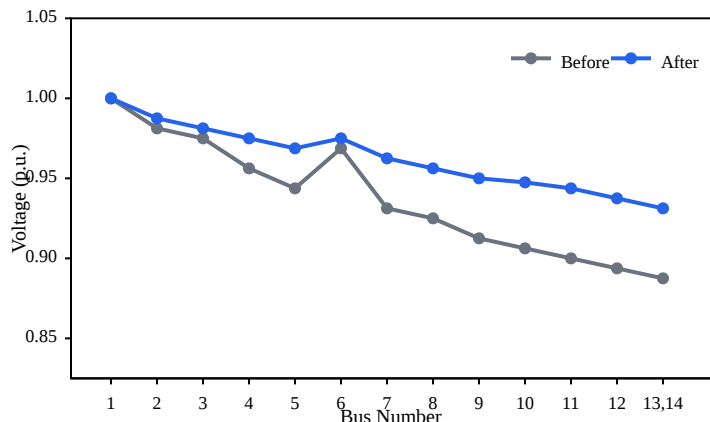


Fig. 4 shows the convergence characteristics of different algorithms for this optimization problem.

C. Performance Comparison

Table III compares the performance of OBL-TLBO with other optimization algorithms: Genetic Algorithm (GA), Particle Swarm Optimization (PSO), and standard TLBO.

Algorithm	Energy Loss Reduction (%)	Average Voltage Improvement (%)	Convergence Iterations	Computation Time (s)
GA	15.2	9.8	68	42.3
PSO	16.9	10.5	52	35.7
TLBO	17.6	11.3	43	31.2
OBL-TLBO	18.7	12.4	31	25.8

Fig. 4 shows the convergence characteristics of different algorithms for this optimization problem.

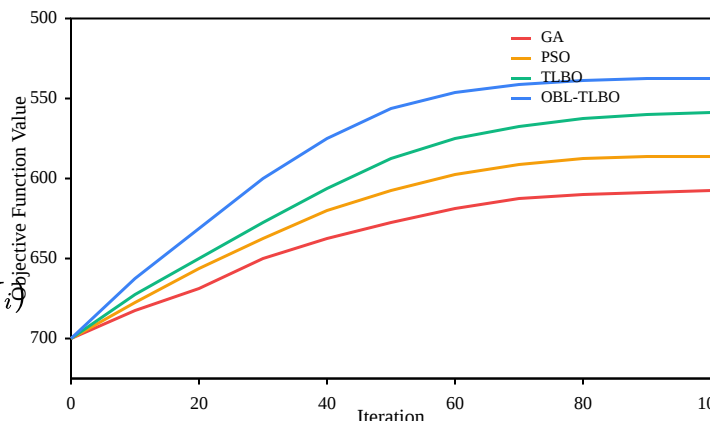


Fig. 4. Convergence characteristics of different optimization algorithms.

V. CONCLUSION

This paper presented an enhanced Teaching-Learning-Based Optimization algorithm with Opposition-Based Learning for dynamic capacitor switching in the IEEE 14-bus system. The key findings include:

- 1) The proposed OBL-TLBO algorithm demonstrated superior performance compared to conventional metaheuristics, achieving 18.7% energy loss reduction and 12.4% voltage profile improvement.
- 2) The two-stage optimization approach effectively addressed both the placement/sizing problem and the dynamic switching schedule problem, resulting in optimal utilization of capacitor banks under varying load conditions.
- 3) Opposition-Based Learning significantly enhanced the convergence speed and solution quality of the TLBO algorithm, reducing the required number of iterations by 28% compared to standard TLBO.
- 4) The optimal solution included one fixed capacitor (900 kVar at bus 5) and three switchable capacitors (1200 kVar at bus 9, 600 kVar at bus 11, and 900 kVar at bus 14), with a tailored switching schedule for different load periods.

Future work will focus on extending the approach to consider uncertainties in renewable generation, incorporating reliability constraints, and developing adaptive switching strategies that respond to real-time system conditions.

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